

The Active Role-Jet Orbit Calculus for Constraint-Surface Roles

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Abstract

This paper develops the foundational calculus of *Directive–Substrate–Product* (DBP) role data on a local three-role relation $F(D, S, P) = 0$. The central object is not a passive permutation of coordinate labels but an *active* family of charts: the same relation re-solved with each coordinate as output, with finite jet data transported through the resulting inverse-function maps. On the regular locus this active transposition is an exact birational representation of S_3 on finite jets, localised at the role denominators. The *Active Role-Jet Orbit Theorem* identifies the resulting orbit as the canonical carrier of local DBP invariants: a functional is a DBP invariant exactly when it factors through the orbit quotient, so invariants are obtained, not postulated. The same construction yields a channel-reduction exact sequence in which a channel account is a gauge- and frame-relative representative over an intrinsic invariant quotient; in the second-order curvature case this is the three-channel Gaussian decomposition $K_G = \kappa_c + \kappa_{\text{int}} + \kappa_s$. From the explicit second-jet transport we obtain the named output-role coupling channels $\Lambda_P, \Lambda_D, \Lambda_S$, the coupling carrier $O = (\kappa_c^{(P)}, \kappa_c^{(D)}, \kappa_c^{(S)})$, and the role anisotropy scalar $\mathcal{A}_c$. A faithfulness corollary shows the carrier map has Jacobian determinant $8 \Lambda_P \Lambda_D \Lambda_S / q_0^6$, so the carrier separates active role orbits off the channel-isotropy loci. Singular points where an output chart cannot be solved are treated as typed strata, not removable cases. Every claim is tagged by certainty tier; the construction is self-contained.

Notation. Throughout, K_G is the Gaussian curvature. For a graph $P = f(D, S)$ we write the first and second jet as $(a, b, A, B, C) = (f_D, f_S, f_{DD}, f_{DS}, f_{SS})$ and set $q_0 = 1 + a^2 + b^2$; for the implicit relation $F = P - f$ one has $g = \nabla F$ and $q_0 = g^\top g$, so the graph normaliser and the implicit gradient norm coincide in the graph gauge. We distinguish two “three-channel” structures and never conflate them: the channel *types* $\kappa_c, \kappa_{\text{int}}, \kappa_s$ (coupling, interaction, self) decompose K_G *within one chart*; the chart-indexed coupling channels $\kappa_c^{(\rho)}$, $\rho \in \{P, D, S\}$, are the coupling type evaluated in the three output-role *charts*, and the anisotropy $\mathcal{A}_c$ is their spread. “The three channels” is never used unqualified.

1 Introduction: active versus passive role data

Let (D, S, P) denote a distinguished ordered triple of local coordinates. DBP studies a local relation among them by comparing the finite jets obtained when the same relation is solved with different coordinates as output. The basic object is therefore neither a labelled triple alone nor a passive permutation of a fixed derivative array; it is an active family of charts related by inverse-function maps.

This distinction is essential. If one merely permutes coordinate labels, a symmetric expression is unchanged for the trivial reason that the indices were renamed; the calculation verifies covariance

but measures no role dependence. Active role transposition is different: it changes the graph representative of the same local relation. The derivative coordinates then transform rationally and need not agree componentwise.

The development below is mechanical. First, the relation is restricted to the regular locus where each output chart exists (§2). Second, finite jets are transported by the active role maps, which realise an exact birational S_3 action (§3). Third, the active orbit is used as the carrier for local DBP invariants (§4); the quotient property of invariants is standard, but the explicit active action, its rational finite-jet closure, its separation from passive covariance, and the induced channel reduction are the content. The remaining sections specialise the channel reduction to curvature (§6), extract the named output-role channels and prove faithfulness of the coupling carrier (§7), record gauge transport (§8), type the singular strata (§9), and delimit scope (§10).

2 Local role data

2.1 Three-role relation

Let $U \subset \mathbb{R}^3$ be open with coordinates (D, S, P) , let $F \in C^{r+1}(U)$, and let $x_0 = (D_0, S_0, P_0) \in U$ satisfy $F(x_0) = 0$. A local graph chart with P as output is written $P = f(D, S)$. The implicit and graph descriptions agree on any chart in which the relevant partial of F is nonzero. We write $j^r f$ for the order- r jet of the graph representative and $j^r F$ for the order- r implicit jet.

2.2 Regular active-role locus

Active role transposition requires the relation to be solvable with each coordinate as output. Define the first-order regular role locus

$$\Omega_1 = \{x \in U : F(x) = 0, F_D(x)F_S(x)F_P(x) \neq 0\},$$

and more generally let Ω_r denote the locus on which all output charts needed for the order- r active role object exist. In graph form $P = f(D, S)$ with $a = f_D$, $b = f_S$, full first-order active recharting is defined where

$$a \neq 0, \quad b \neq 0.$$

These are the denominators of the inverse-function maps, not technical decorations: a point outside the active-role locus is a singular stratum for the requested role object (§9).

3 Active role transposition: the birational S_3

3.1 First-order action

For a graph chart $P = f(D, S)$ with $a = f_D$, $b = f_S$, the input swap $s = (D \ S)$ and the active transposition $t = (D \ P)$ act by

$$s(a, b) = (b, a), \quad t(a, b) = \left(\frac{1}{a}, -\frac{b}{a}\right),$$

obtained by re-solving $F = 0$ for D as output. These satisfy

$$s^2 = t^2 = e, \quad (st)^3 = e,$$

so active role transposition is an exact birational representation of S_3 on the regular jet locus, localised at the role denominators. [proven]

3.2 Second-order generator

With $A = f_{DD}$, $B = f_{DS}$, $C = f_{SS}$, the input swap acts by

$$s(a, b, A, B, C) = (b, a, C, B, A).$$

Under $t = (D \ P)$ the chart $D = q(P, S)$ has

$$q_P = \frac{1}{a}, \quad q_S = -\frac{b}{a}, \quad q_{PP} = -\frac{A}{a^3}, \quad q_{PS} = \frac{Ab - aB}{a^3}, \quad q_{SS} = \frac{-Ab^2 + 2abB - a^2C}{a^3},$$

so that

$$t(a, b, A, B, C) = \left(\frac{1}{a}, -\frac{b}{a}, -\frac{A}{a^3}, \frac{Ab - aB}{a^3}, \frac{-Ab^2 + 2abB - a^2C}{a^3} \right).$$

The maps again satisfy the S_3 relations on the regular locus. The mixed second-order numerator $Ab - aB$ appearing in q_{PS} is the source of the directive coupling channel of §7. Higher-order generators follow by the finite inverse-function theorem and the chain rule; at each finite order they are rational in the original jet, localised at the first-order role denominators (Appendix A). [proven]

4 The Active Role-Jet Orbit Theorem

Definition 1 (jet carrier). Let $J_r(x_0)$ be the order- r jet of the chosen local graph representative at x_0 . The active order- r role-jet orbit is $\mathfrak{R}_r(x_0) = G \cdot J_r(x_0)$, where the action is active role recharting. Retaining the coordinate values gives the value-plus-jet orbit $\mathfrak{R}_r^+(x_0) = G \cdot (x_0, J_r(x_0))$.

Definition 2 (local DBP invariant). Let X_r be the order- r jet space on the regular role locus and Y any target set. A functional $\phi : X_r \rightarrow Y$ is an order- r DBP invariant if $\phi(\mathcal{T}_\sigma \xi) = \phi(\xi)$ for all $\sigma \in S_3$ wherever both sides are defined. For value-plus-jet invariants, replace X_r by X_r^+ .

Theorem 1 (Active Role-Jet Orbit Theorem). *On the regular active-role locus, let $G = S_3$ act on X_r by the active finite-jet recharting maps of §3, and let $\pi : X_r \rightarrow X_r/G$ be the orbit quotient. Then $\phi : X_r \rightarrow Y$ is a local order- r DBP invariant if and only if there is a unique $\psi : X_r/G \rightarrow Y$ with $\phi = \psi \circ \pi$. Equivalently, the value of any order- r DBP invariant at x_0 is a function of the active orbit $\mathfrak{R}_r(x_0) = G \cdot J_r(x_0)$. The same holds for value-plus-jet functionals with X_r^+ and $\mathfrak{R}_r^+(x_0)$. [proven]*

Proof. The maps $\mathcal{T}_\sigma$ define an action of S_3 on the regular active-role locus. By definition a DBP invariant is constant on its orbits. The quotient map π identifies exactly the elements lying in the same orbit; hence a functional is constant on active role orbits iff it factors uniquely through the quotient. The value-plus-jet statement is identical with X_r^+ . The quotient step is formal; the mathematical content is the prior construction of the active finite-jet action, in which the representatives in different output roles are rational inverse-function transforms, not passive relabellings. $\square$

5 Consequences of the orbit carrier

5.1 Completeness criterion

A finite tuple K_r is a complete order- r DBP kernel only if it separates the same active role orbits as the quotient map:

$$K_r(\xi_1) = K_r(\xi_2) \iff \pi(\xi_1) = \pi(\xi_2)$$

on the intended domain. A preferred kernel cannot be assumed complete merely because its entries are invariant; completeness requires orbit separation. [proven] This criterion is the test applied to the coupling carrier in §7.3.

5.2 Exact rational closure

If the original finite jet has entries in a field $\mathbb{K}$ and all active role denominators are nonzero, every active role-transposed jet has entries in the localisation obtained by inverting those denominators. In particular

$$J_r(x_0) \in \mathbb{Q} \implies G \cdot J_r(x_0) \subset \mathbb{Q}[\Delta^{-1}],$$

where Δ is the product of the role denominators used by the active charts. [proven] Thus for exact rational input jets, active role transport is an exact rational operation on the regular locus; this is the statement that licenses exact- $\mathbb{Q}$ computation of every quantity below.

5.3 Descent classes

Invariants separate by the orbit data they require. *Value-only* invariants descend to the quotient of coordinate values alone; *jet-only* coupling invariants descend to the quotient of the jet without values; *mixed* value-jet invariants require the value-plus-jet orbit $\mathfrak{R}_r^+$. The curvature channels of §6 are jet-only; the power-law degree spectra of Appendix B are value-only in log-role form. [proven]

6 Channel reduction

6.1 Abstract reduction law

Let C_r be a channel-account space at order r and let $\Sigma : C_r \rightarrow I_r$ be a surjective reduction onto an invariant range. Then

$$0 \longrightarrow \ker \Sigma \longrightarrow C_r \xrightarrow{\Sigma} I_r \longrightarrow 0$$

is exact, and $C_r / \ker \Sigma \cong I_r$ is the invariant. The fibre over a fixed value of I_r is the channel account: a representative may move inside the fibre under chart or gauge change while the reduced invariant is fixed. [proven]

6.2 Curvature-channel projection

For a hypersurface relation $F = 0$ at a regular point set $g = \nabla F$, $H = \nabla^2 F$, $q = g^\top g \neq 0$. In the distinguished coordinate basis split

$$H = H_c + H_s, \quad H_s = \text{diag}(H), \quad H_c = H - H_s.$$

For $1 \leq r \leq n - 1$ define the channel-density polynomial

$$\widehat{\mathcal{C}}_r(t, u) = (-1)^{r+1} \sum_{|I|=r+1} \det \begin{pmatrix} 0 & g_I^\top \\ g_I & (tH_c + uH_s)_I \end{pmatrix} = \sum_{p+q=r} t^p u^q \widehat{\kappa}_{r;p,q}.$$

The normalised channel curvatures are $\kappa_{r;p,q} = \widehat{\kappa}_{r;p,q} / q^{(r+2)/2}$, and the reduced elementary curvature invariant is

$$\sigma_r = \sum_{p+q=r} \kappa_{r;p,q} = \frac{\widehat{\mathcal{C}}_r(1, 1)}{q^{(r+2)/2}}.$$

Thus the reduction is summation, $\Sigma : C_r \mapsto \sum_{p+q=r} \kappa_{r;p,q} = \sigma_r$: the channel vector is the representative, its sum is the invariant. [proven]

6.3 The three-variable Gaussian case

For $n = 3$, $r = 2$ the sum has three terms; writing $\kappa_c = \kappa_{2;2,0}$ (the t^2 , pure-coupling term), $\kappa_{\text{int}} = \kappa_{2;1,1}$ (the tu cross term), $\kappa_s = \kappa_{2;0,2}$ (the u^2 , pure-self term),

$$C_2 = (\kappa_c, \kappa_{\text{int}}, \kappa_s), \quad K_G = \kappa_c + \kappa_{\text{int}} + \kappa_s.$$

The zero-sum transport plane is $\ker \Sigma = \{(u, v, w) : u + v + w = 0\}$: a gauge or chart change may move the channel triple within the affine plane over K_G , but cannot change the reduced Gaussian curvature. [proven]

6.4 Graph-channel formula

For any local graph $z = h(x, y)$ with $\alpha = h_x$, $\beta = h_y$, $L = h_{xx}$, $M = h_{xy}$, $N = h_{yy}$ and $Q = 1 + \alpha^2 + \beta^2$,

$$\kappa_c = -\frac{M^2}{Q^2}, \quad \kappa_s = \frac{LN}{Q^2}, \quad \kappa_{\text{int}} = 0, \quad K_G = \frac{LN - M^2}{Q^2}.$$

Every graph-normalised output chart therefore has zero graph-channel interaction; interaction reappears in non-graph implicit gauges or under gauge transport between graph sections (§8). [proven]

6.5 Curved-ambient extension

The channel reduction extends to a curved ambient space through the Gauss equation, which reads schematically

$$R^\Sigma = \bar{R}^T + S \wedge S,$$

where R^Σ is the intrinsic curvature of the role surface, $\bar{R}^T$ is the tangential part of the ambient background curvature, and the extrinsic term is quadratic in the shape operator S . Applying the DBP split to the shape operator, $S = S_c + S_s$, the wedge is bilinear, so

$$S \wedge S = S_c \wedge S_c + (S_c \wedge S_s + S_s \wedge S_c) + S_s \wedge S_s = R_{cc} + R_{\text{int}} + R_{ss} :$$

the extrinsic contribution channelises into pure-coupling, interaction, and pure-self parts exactly as in the flat case. Hence

$$\text{intrinsic invariant} = \text{ambient input} + \text{extrinsic DBP-channel account}.$$

[proven] The ambient term $\bar{R}^T$ is *not* forced into the same self/coupling split, and this is a boundary of the construction with a named cause rather than a gap in it: the extrinsic term is a bilinear function of the shape operator and therefore splits along the role frame, whereas the ambient curvature is not a bilinear function of the same role data, so no channel split is available for it unless the application supplies a separate role frame for the ambient geometry itself. [proven]

7 Output-role coupling channels, anisotropy, and faithfulness

7.1 The three output-role charts

For $P = f(D, S)$ with $q_0 = 1 + a^2 + b^2$, the three output-role charts carry the channel quadruples $(K_G, \kappa_c, \kappa_s, \kappa_{\text{int}})$

$$\begin{aligned} C_P &= \left(K, -\frac{B^2}{q_0^2}, \frac{AC}{q_0^2}, 0 \right), \\ C_D &= \left(K, -\frac{(Ab-aB)^2}{a^2 q_0^2}, \frac{A(Ab^2-2abB+a^2C)}{a^2 q_0^2}, 0 \right), \\ C_S &= \left(K, -\frac{(Ca-bB)^2}{b^2 q_0^2}, \frac{C(Ca^2-2abB+b^2A)}{b^2 q_0^2}, 0 \right), \end{aligned}$$

with the common Gaussian curvature $K = (AC - B^2)/q_0^2$. The scalar output-role curvature spectrum is $\text{OutputRoleCurvSpec}(f) = \{C_P, C_D, C_S\}$; retaining ordered input labels gives the honest sixfold labelled role-jet curvature object. [symbolic]

7.2 Named coupling channels and the anisotropy scalar

Define the output-role mixed numerators

$$\Lambda_P = B, \quad \Lambda_D = \frac{Ab - aB}{a}, \quad \Lambda_S = \frac{Ca - bB}{b},$$

so that the chart-indexed coupling channels are $\kappa_c^{(\rho)} = -\Lambda_\rho^2/q_0^2$ for $\rho \in \{P, D, S\}$. The coupling carrier is $O = (\kappa_c^{(P)}, \kappa_c^{(D)}, \kappa_c^{(S)})$. The role anisotropy scalar is

$$\mathcal{A}_c = (\kappa_c^{(P)} - \kappa_c^{(D)})^2 + (\kappa_c^{(D)} - \kappa_c^{(S)})^2 + (\kappa_c^{(S)} - \kappa_c^{(P)})^2,$$

which vanishes exactly when all three output-role coupling channels agree. [proven] Under the permutation action of the role S_3 on $O \in \mathbb{R}^3$ (permuting the three components), $\mathbb{R}^3$ splits as trivial $\oplus$ standard, and a direct computation gives $\mathcal{A}_c = 3\|P_2 O\|^2$, the squared norm of the standard-irrep part: $\mathcal{A}_c$ is the intrinsic isotropy defect of the coupling carrier. [proven] A product-role excess diagnostic is $E_P = \kappa_c^{(P)} - \frac{1}{2}(\kappa_c^{(D)} + \kappa_c^{(S)})$, with analogues for D and S .

7.3 Faithfulness of the coupling carrier

The completeness criterion (§5) asks whether the carrier separates active role orbits. The following corollary answers this locally and ties the answer directly to the channels.

Corollary 1 (faithful coupling carrier). *On the regular role locus, the coupling-carrier map of the second jet,*

$$(A, B, C) \mapsto O = (\kappa_c^{(P)}, \kappa_c^{(D)}, \kappa_c^{(S)}),$$

has Jacobian determinant

$$\det \frac{\partial O}{\partial(A, B, C)} = \frac{8 \Lambda_P \Lambda_D \Lambda_S}{q_0^6}.$$

Hence the carrier map has rank 3 wherever $\Lambda_P \Lambda_D \Lambda_S \neq 0$; it is then a local diffeomorphism and therefore separates active role orbits locally, so by the completeness criterion O is a complete order-2 coupling kernel off the channel-isotropy loci. The carrier degenerates exactly where a coupling numerator vanishes, i.e. on the loci where the corresponding channel is isotropic. [symbolic]

Proof. With $\kappa_c^{(P)} = -B^2/q_0^2$, $\kappa_c^{(D)} = -(Ab - aB)^2/(a^2q_0^2)$, $\kappa_c^{(S)} = -(Ca - bB)^2/(b^2q_0^2)$ and q_0 independent of (A, B, C) , the Jacobian has a single nonzero entry in the first column (from $\kappa_c^{(D)}$) and in the third column (from $\kappa_c^{(S)}$). Expanding and simplifying gives the stated determinant; the factorisation is $8\Lambda_P\Lambda_D\Lambda_S/q_0^6$ on substituting $Ab - aB = a\Lambda_D$, $Ca - bB = b\Lambda_S$, $B = \Lambda_P$. Full column rank is equivalent to a nonzero determinant, and a nonzero Jacobian gives a local diffeomorphism, hence local orbit separation; the completeness criterion then applies. The identity was confirmed in exact symbolic arithmetic. $\square$

Thus the carrier is faithful precisely off the isotropy loci, the same loci on which $\mathcal{A}_c$ records a collapsing coupling channel; faithfulness and anisotropy are two readings of the channel numerators.

8 Gauge transport

The reduction Σ of §6 has a nontrivial kernel: the zero-sum plane for the Gaussian case. A channel representative is therefore gauge- and frame-relative, free to move inside the fibre over the fixed invariant. Concretely, while the graph-channel formula sets $\kappa_{\text{int}} = 0$ in any graph gauge, transport between graph sections of the same relation, or passage to a non-graph implicit gauge, moves weight between κ_c , κ_{int} and κ_s along $\ker \Sigma$, with $K_G = \sigma_2$ held fixed. The interaction channel is therefore live precisely when the working gauge is not a graph gauge; the keystone of Appendix B exhibits an implicit relation whose entire curvature is carried by κ_{int} . [symbolic]

9 Singular strata

The active maps are defined only where their denominators are nonzero. Where a required denominator vanishes the corresponding output chart does not exist, and the correct object is a typed singular stratum, not a coerced scalar. Typical cases:

1. a coordinate cannot be solved locally as output;
2. a projective normal coordinate vanishes;
3. a curvature denominator degenerates;
4. a requested channel representative is not defined in the chosen frame.

A singular stratum may encode a symmetry, a boundary, a chart failure, or a refusal condition; in all cases it is part of the role-jet structure rather than an exception to be removed. [proven]

10 Scope and summary

The theorem is local and finite-order: it describes order- r jets near a regular point, and global statements require additional gluing data across charts. It is frame-relative: the channel split is made in the distinguished coordinate frame, so the reduced invariant may be intrinsic while a channel account is basis-, chart-, or gauge-relative. The theorem does not assert that every useful scalar is invariant; it gives a test, namely that a scalar is a DBP invariant exactly when it is constant on the active role orbits of the relevant datum.

In compact form:

DBP invariant data is carried by the active role-jet orbit;
 DBP channel data is the fibre over its invariant reduction.

A Higher-order transposition

The active transposition $t = (D \ P)$ on the order- r jet is obtained from $D = q(P, S)$ by the finite inverse-function theorem: each q -derivative is a rational function of the original f -jet with denominators a power of $a = f_D$. The first two orders are given in §3; the third-order reach proceeds identically, differentiating the relations $q_P = 1/a$, $q_S = -b/a$ once more and substituting, so that every order- r component lies in $\mathbb{Q}[\text{jet}, a^{-1}]$ in agreement with §5. The input swap acts at every order by the index transposition $D \leftrightarrow S$, e.g. $s(a, b, A, B, C) = (b, a, C, B, A)$ at order 2. [proven]

B Worked examples

B.1 Power-law charts: value-only role spectra

For a power-law coupling $P = D^m S^n$ pass to log coordinates, $\log P = m \log D + n \log S$, with exponent normal $\ell = (-m, -n, 1)$. Role transposition permutes the entries of this projective normal, giving the exponent-pair orbit

$$\mathcal{E}(m, n) = \left\{ (m, n), (n, m), \left(\frac{1}{m}, -\frac{n}{m}\right), \left(-\frac{n}{m}, \frac{1}{m}\right), \left(-\frac{m}{n}, \frac{1}{n}\right), \left(\frac{1}{n}, -\frac{m}{n}\right) \right\}$$

and the role-degree spectrum

$$\text{DegSpec}(m, n) = \left\{ m+n, m+n, \frac{1-n}{m}, \frac{1-n}{m}, \frac{1-m}{n}, \frac{1-m}{n} \right\}.$$

For ordinary multiplication $P = DS$, $(m, n) = (1, 1)$ gives $\text{DegSpec}(1, 1) = \{2, 2, 0, 0, 0, 0\}$: the forward chart has degree 2 while inverse charts $D = P/S$ have degree 0 in log-role form. For the cubic mixed power $P = D^2 S$, $(m, n) = (2, 1)$ gives $\text{DegSpec}(2, 1) = \{3, 3, 0, 0, -1, -1\}$, distinguishing it from ordinary multiplication though both are monomial. Scale-free exponent invariants are

$$I_1^\ell = \frac{(1-m-n)(mn-m-n)}{mn}, \quad I_2^\ell = \frac{(1-m-n)^3}{mn},$$

with exponent discriminant $\Delta_\ell = (m-n)^2(m+1)^2(n+1)^2$. [symbolic] At multiplication the coupling channel is flat, so $\mathcal{A}_c = 0$; the keystone below is nowhere flat.

B.2 The keystone: interaction-only curvature

Consider the implicit relation

$$F = x_1^2 + x_1 x_2 + x_3^2 - 3 \quad \text{at} \quad p = (1, 1, 1).$$

Then $g = (3, 1, 2)$, $q = 14$, and

$$H = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

The channel-density polynomial of §6 gives

$$\kappa_c = -\frac{1}{49}, \quad \kappa_s = \frac{1}{49}, \quad \kappa_{\text{int}} = -\frac{3}{49}, \quad \text{so} \quad K_G = \kappa_c + \kappa_{\text{int}} + \kappa_s = -\frac{3}{49}.$$

The coupling and self channels cancel exactly and the surviving curvature is interaction curvature. Any non-negative scalar proxy, or any single-channel account that omits κ_{int} , cannot represent this object; the keystone is the discriminating witness that the three-channel decomposition is not reducible to a single number. [symbolic] (Channels confirmed in exact arithmetic against the direct Gauss–Kronecker value $K_G = -3/49$.)

B.3 Gauge transport on the keystone

A defining-function gauge $\tilde{F} = \mu F$ with $\mu(p) = 1$ and $a = \nabla \log \mu(p)$ fixes the gradient and shifts the Hessian by

$$\tilde{g} = g, \quad \tilde{H} = H + ga^\top + ag^\top.$$

Writing the gauge vector as $a = (u, v, w)$ on the keystone surface above, the channels transport by the exact rational laws

$$\begin{aligned} \kappa_c(u, v, w) &= \frac{1}{49}(12uv + 6uw + 18vw + 6w - 1), \\ \kappa_s(u, v, w) &= \frac{1}{49}(12uv + 6uw + 3u + 18vw + 13v + 2w + 1), \\ \kappa_{\text{int}}(u, v, w) &= -\frac{1}{49}(24uv + 12uw + 3u + 36vw + 13v + 8w + 3), \end{aligned}$$

and for every rational gauge vector

$$\kappa_c(u, v, w) + \kappa_s(u, v, w) + \kappa_{\text{int}}(u, v, w) = -\frac{3}{49}.$$

The channel triple therefore moves inside the affine zero-sum plane $\Delta\kappa_c + \Delta\kappa_s + \Delta\kappa_{\text{int}} = 0$ while the Gaussian curvature is held fixed, concretely realising the kernel of the reduction Σ (§6) and the gauge transport of §8. The pure coupling channel is pinned, $\kappa_c = -\frac{1}{49}$, exactly on the locus

$$w(u + 3v + 1) + 2uv = 0,$$

so channel pinning is a structural condition on the gauge, not an accident: a generic gauge moves every channel, a special gauge fixes a chosen one. [symbolic]

C Application patterns and working checklist

The calculus is applied by supplying a role assignment and a relation and then reading the orbit and channel account at the order the question requires.

- **Generic relation.** Assign D (directive, policy, or design pressure), S (substrate, state, or resource), P (product, output, or reading), and write $P = f(D, S)$ or $F(D, S, P) = 0$. The role orbit records what survives re-solving from each role; the channel account records how the invariant is distributed in the chosen frame.
- **Sensitivity.** If only first order matters, take $(a, b) = (f_D, f_S)$ and its active orbit; projective role invariants diagnose role symmetry, stabilisers, and inverse-role boundaries.
- **Curvature.** If second-order interaction matters, take (a, b, A, B, C) and read the output-role spectra C_P, C_D, C_S , with K_G the reduced invariant and the channel triples the explanatory representatives.
- **Route/account.** If several routes reach the same reading, treat route choice as a role-structured relation; the invariant quotient is route-independent, and a loop holonomy is a coupling diagnostic that vanishes when routes combine additively and is nonzero when they genuinely interact.
- **Monomial scaling.** For $P = D^m S^n$, use the exponent orbit and degree spectrum to detect when a monomial that is simple in one role becomes inverse, neutral, or singular in another (Appendix B).

A working checklist for a new application:

	Object	Question
1	Roles (D, S, P)	What are the directive, substrate, product roles?
2	Relation	Is the local law $P = f(D, S)$ or $F(D, S, P) = 0$?
3	Regular locus	Which role denominators must be nonzero?
4	Jet order r	Sensitivity, curvature, or higher holonomy?
5	Active role orbit	What are the transformed role charts?
6	Candidate invariant	Does it factor through the active role quotient?
7	Channel account	Which explanatory channels are retained?
8	Reduction map	What is the invariant quotient of the account?
9	Kernel / fibre	What moves under chart or gauge while the invariant stays fixed?
10	Singular strata	What must refuse, rechart, or be typed at a boundary?

Provenance of claims

The S_3 relations and the first- and second-order transposition formulas are explicit rational identities ([proven], symbolic-CAS confirmable). Theorem 1, the completeness criterion, the exact rational closure, the channel-reduction exact sequence, the Gaussian decomposition $K_G = \kappa_c + \kappa_{\text{int}} + \kappa_s$, the graph-channel formula, the bilinear curved-ambient split $S \wedge S = R_{cc} + R_{\text{int}} + R_{ss}$ with its named ambient boundary, and the definitions and identities of the named channels and $\mathcal{A}_c = 3\|P_2O\|^2$ are [proven]. Corollary 1 (the carrier Jacobian determinant $8\Lambda_P\Lambda_D\Lambda_S/q_0^6$ and rank 3), the keystone channel values, and the keystone gauge-transport laws were established and confirmed in exact symbolic arithmetic, [symbolic]. The construction is self-contained: no external named theorem is load-bearing, and every transported quantity is exact- $\mathbb{Q}$ on the regular locus by §5.